

Arduino & SG90 Servo Motor Control

Building Smart Gates & Precise Angular Control | Student Lesson & Hands-On Worksheet

Required Materials

- Arduino UNO Board
- SG90 Micro Servo Motor
- Potentiometer (for precise angle control input)
- Jumper Wires (Male-to-Male / Male-to-Female)

1. Introduction: How Does a Toll Gate Lift Up?

Have you ever noticed how automatic toll barriers at FASTag gates or residential security gates lift up to let cars pass? The barrier arm always lifts to an exact 90° angle, stops firmly in mid-air, and then lowers back down to 0° once the vehicle passes.

How does a machine achieve such precise positioning every single time? The answer lies in a special component called a **Servo Motor** controlled by a micro-controller like the Arduino UNO.

2. What is a Servo Motor?

A servo motor is a rotary actuator that allows for precise control of angular position, velocity, and acceleration. Unlike standard motors, it knows its exact position and holds it firmly.

Servo Motor	Standard DC Motor
Moves to a specific target angle (e.g., 0° to 180°).	Spins continuously in a circle as long as power is applied.
Holds its position firmly against external resistance.	Cannot hold a precise position or fixed angle.
Ideal for robotics, smart gates, steering mechanisms.	Ideal for wheels, fans, and simple spinning mechanisms.

3. Meet the SG90 Servo Motor & Wiring Guide

The SG90 is a lightweight, high-output micro servo motor popular in educational electronics. It operates on 3 wires:

Wire Color	Function	Arduino UNO Connection	Mnemonic / Memory Trick
Red	VCC (Power Supply)	5V Pin	Red = Power / Energy
Brown / Black	GND (Ground / Earth)	GND Pin	Brown/Black = Earth / Ground
Yellow / Orange	Signal (PWM Control)	Digital Pin 9 (PWM)	Yellow = Traffic Signal (Control)

Spot the Wiring Mistake!

Always double-check your connections before powering on your Arduino! Common mistakes include swapping the Signal and VCC lines or plugging the power line into a 3.3V pin instead of 5V. Incorrect wiring can prevent the motor from moving or cause erratic behavior.

4. Step-by-Step Practical Implementation

Follow these steps using your materials (**Arduino, SG90 Servo, Potentiometer, and Jumper Wires**) to complete the workstation lab:

1. Step 1: Circuit Assembly (Basic Servo)

Connect the SG90 Servo Motor to the Arduino UNO using jumper wires: Red to 5V, Brown/Black to GND, and Yellow/Orange to Digital Pin 9.

2. Step 2: Potentiometer Integration (Manual Angular Control)

Connect the Potentiometer to control the servo position dynamically:

- Connect the left pin of the Potentiometer to GND using a jumper wire.
- Connect the right pin of the Potentiometer to 5V using a jumper wire.
- Connect the middle pin (wiper) of the Potentiometer to Analog Pin A0 on the Arduino.

3. Step 3: Basic Code & Servo Library Setup

Upload the following fundamental code to verify basic servo control:

```
// Step 1: Include the Servo Library
#include <Servo.h>

// Step 2: Create a Servo Object
Servo myServo;

void setup() {
  // Step 3: Attach the signal pin to Digital PWM Pin 9
  myServo.attach(9);
}

void loop() {
  // Step 4: Command servo to move to 90 degrees
  myServo.write(90);
  delay(1000); // Wait 1 second for the motor to reach position
}
```

4. Step 4: Interactive Potentiometer Control Code

Upload the sketch below to read the potentiometer value from A0 (0–1023), map it to servo angles (**0°–180°**), and drive the servo accordingly:

```
#include <Servo.h>

Servo myServo;
int potPin = A0; // Potentiometer middle pin
int potVal;      // Variable to store raw analog reading
int angle;       // Variable to store converted angle

void setup() {
  myServo.attach(9);
}

void loop() {
  potVal = analogRead(potPin); // Read value from 0 to 1023
  angle = map(potVal, 0, 1023, 0, 180); // Map to angle between 0 and 180 degrees
  myServo.write(angle); // Move servo to target angle
  delay(15); // Smooth transition delay
}
```

5. Step 5: Automated Smooth Sweep Demonstration

To simulate a smooth continuous sweep without manual potentiometer control, test the following loop logic:

```
void loop() {
  // Sweep from 0 to 180 degrees
  for (int pos = 0; pos <= 180; pos += 1) {
    myServo.write(pos);
    delay(15); // Wait 15ms for smooth motion
  }
  // Sweep back from 180 to 0 degrees
  for (int pos = 180; pos >= 0; pos -= 1) {
    myServo.write(pos);
    delay(15);
  }
}
```

5. Real-World Applications in India & Beyond

- **FASTag Toll Barriers:** Lifts the barrier arm exactly **90°** when a valid tag is scanned and closes it after the vehicle passes.
- **Residential Society Boom Barriers:** Automates security entry and exit gates for gated communities.
- **ISRO Space Technology:** Solar panels on satellites and rover robotic arms rely on precise servo actuators to deploy and position instruments accurately in space.

6. DIY Hands-On Challenge: Build Your Smart Gate Arm!

Challenge Materials: SG90 Servo Motor, Arduino UNO, Potentiometer, Jumper Wires, Popsicle Stick, Double-sided tape.

1. Fix a popsicle stick to the plastic servo horn/arm using double-sided tape or Blu-tack.
2. Connect the servo and potentiometer to the Arduino according to the wiring steps outlined above.
3. Write/Modify a program that opens the barrier to **90°** when commanded (or via potentiometer knob turn), holds it open for 3 seconds, and then closes it back to **0°**.

7. Practice Questions & Assessment

Question 1 — Multiple Choice

How many wires does an SG90 servo motor have, and what are their specific functions?

- A) 2 wires: Power (+) and Ground (-)
- B) 3 wires: Red (Power 5V), Brown/Black (Ground), Yellow/Orange (Signal)
- C) 4 wires: VCC, GND, Step, and Direction
- D) 3 wires: Red (Signal), Black (5V Power), Yellow (Ground)

Question 2 — Conceptual Comparison

Explain the main technical difference between a standard DC motor and a Servo motor. In what practical scenario would you choose a servo motor over a DC motor?

Question 3 — Arduino Programming

Which Arduino header file must be included at the top of your sketch to control a servo motor, and what method/function connects the servo object to a specific digital pin?

Question 4 — Code Analysis

Look at the following line of code: `myServo.write(90);`. What physical action does this command cause the servo motor to perform? How long will it stay in that position if no other commands are given?

Question 5 — Troubleshooting & Wiring

A student connects an SG90 servo motor and a potentiometer to an Arduino UNO, but when they turn the potentiometer knob, the motor makes a buzzing sound and does not move properly. List two potential wiring or power issues that could cause this problem.

Question 6 — Real-World Engineering Challenge

Design an algorithmic flow (or write pseudo-code) for an automated FASTag toll gate system controlled via potentiometer input or automated sensors. Describe how the gate should behave from the moment a vehicle approaches to when it drives through and leaves. Mention the servo angles used for opening and closing.